

Supporting Information for

Optimal plant water use strategies explain soil moisture variability

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This PDF file includes:

- **Methods S1** Data selection
- **Methods S2** Model parameterization
- **Table S1** Selected FLUXNET2015 sites characteristics and references.
- **Figure S1** Evapotranspiration partitioning versus leaf area index.
- **Figure S2** Effect of soil suitability on water use strategies.
- **Figure S3** Effect of soil-root transport capacity on water use strategies.
- **Figure S4** Effect of soil texture on water use strategies.
- **Figure S5** Effect of root area index on water use strategies.
- **Figure S6** Same as Figure 9 but considering static versus dynamics water stress.
- **Figure S7** Relation between plant water flux control and hydraulic risk tolerance.
- **References**

Methods S1 Data selection

We use daily observations from 40 sites in the FLUXNET2015 dataset (Pastorello et al., 2020) and identify site characteristics from available metadata and references (Table S1). We select sites and quality check records to best meet soil water balance model assumptions and only analyze growing season data. The model requires rainfall, net radiation, and air temperature climate inputs. We evaluate model results against soil moisture observations as well as independent (not used in parameter inference) total evapotranspiration and gross primary productivity fluxes measured by eddy-covariance. For each site coordinates, we also extract *LAI* estimates from the Moderate Resolution Imaging Spectroradiometer (MODIS product MCD15A3H-006) with a spatial resolution of 500 m and temporal resolution of 4 days (Myneni et al., 2015).

Selected sites meet the following criteria: soil moisture measurements between 10 and 50 cm depth (we use the deepest available sensor); no signs of groundwater influence on soil water balance (median growing season soil saturation < 0.8); peak growing season *LAI* > 0.5 ; > 300 days of growing seasons data over several years. The growing season period at each site is taken as months during which *LAI* is at least 50% of peak *LAI* (90th percentile of annual pattern) and average air temperature $> 2^{\circ}\text{C}$. We screen soil moisture time series and exclude periods of record with extreme discontinuities that may have been caused by sensor replacements and changes in soil porosity, and years with $> 20\%$ data gaps during the growing season. We consider only cloudless MODIS data flagged as high-quality between 2003-2013, then interpolate and average *LAI* values by day of year, using a 30-day convolution to obtain a smooth annual pattern.

Methods S2 Model parameterization

To explore the role of the Π groups theoretically, we define baseline parameter values for a temperate broadleaf forest (Table 1). To apply the framework at the selected 40 ecosystems, we prescribe site-specific soil texture, vegetation structure, and climate parameters derived from data and metadata (Table S1) and estimate the four unknown plant trait values ($\psi_{g,50}$, $\psi_{x,50}$, $k_{x,max}$, RAI) through Bayesian model inversion.

We first infer parameter values that best-fit soil moisture observations at each study site (data-driven criterion). We use the Bayes theorem to relate theoretical (Eq. 8) and empirical (derived from data) $p(s)$ and estimate plausible distributions of four plant traits ($\psi_{g,50}$, $\psi_{x,50}$, $k_{x,max}$, RAI) and associated Π groups and ecohydrological parameters (f_{ww} , s^* , s_w) using a Metropolis-Hastings Markov chain Monte Carlo algorithm – see Bassiouni *et al.* (2018, 2020) for details, justifications and limitations of this method. Here, the algorithm is adapted for this study’s improved soil water balance, which explicitly partitions evapotranspiration and is eco-physiologically constrained by plant traits represented in Π groups.

We assume uninformed prior knowledge of plant traits and consider uniform distributions over broad ranges reported in the literature (Jackson *et al.*, 1997; Manzoni *et al.*, 2013): $\psi_{g,50}$ and $\psi_{x,50}$ [-10, -0.1] (MPa); $k_{x,max}$ [0.01, 2] ($10^{-3} \text{ kg}^{-1} \text{ m}^{-1} \text{ s}^{-1} \text{ MPa}^{-1}$); RAI [1, 80] ($\text{m}^2 \text{ m}^{-2}$). We repeat the algorithm three times with 20,000 steps (10,000 step burn-in period) and evaluate convergence from the 3 simulations using the Gelman-Rubin diagnostic (Gelman and Rubin, 1992). We ensure physically meaningful results by rejecting samples that do not meet the following constraints: $\psi_{x,50} < \psi_{g,50}$; long-term average evapotranspiration lower than rainfall; leaf water potential at s^* less negative than the leaf water potential threshold that produces maximum transpiration. Finally, we take the median values resulting from the 30,000 combined converging parameters estimates.

We then independently infer parameter values that lead to maximum plant water use performance at each site (optimality-based criterion). We evaluate the same parameter space with the same algorithm but without soil moisture observations and changing the likelihood function to maximize stress-weighted transpiration (ϵ , Eq. 13).

Table S1 Selected FLUXNET2015 sites characteristics and references.

Site ID	IGBP	Climate	DD	Soil	n	Z_m	E_0	α	λ	h_c	LAI	Reference	Data DOI
AT-Neu	GRA	Te	1068	3	0.46	0.1	1.7	6.4	0.33	0.2	1.73	(Wohlfahrt et al., 2008)	https://doi.org/10.18140/FLX/1440121
BR-Sa3	EBF	Tr	1050	9	0.46	0.15	3.8	10.1	0.39	27.5	2.81	(Asner et al., 2004)	https://doi.org/10.18140/FLX/1440033
CA-Oas	DBF	Bo	2125	3	0.45	0.1	2.6	6.9	0.29	21.0	1.91	(Barr et al., 2004)	https://doi.org/10.18140/FLX/1440043
CA-Obs	ENF	Bo	1342	3	0.33	0.1	2.8	6.3	0.36	11.0	1.58	(Bergeron et al., 2007)	https://doi.org/10.18140/FLX/1440044
CA-SF1	ENF	Bo	549	3	0.44	0.3	2.8	8.5	0.25	6.0	1.88	(Mkhabela et al., 2009)	https://doi.org/10.18140/FLX/1440046
CA-SF2	ENF	Bo	366	3	0.27	0.15	2.9	6.2	0.26	4.0	2.33	(Mkhabela et al., 2009)	https://doi.org/10.18140/FLX/1440047
CA-SF3	OSH	Bo	481	3	0.35	0.15	2.8	6.1	0.36	18.0	1.68	(Mkhabela et al., 2009)	https://doi.org/10.18140/FLX/1440048
CH-Lae	MF	Te	1712	6	0.31	0.2	3.5	4.6	0.80	37.0	2.24	(Etzold et al., 2011)	https://doi.org/10.18140/FLX/1440134
CN-Cng	GRA	Te	488	9	0.82	0.1	3.0	9.9	0.16	0.6	0.93	-	https://doi.org/10.18140/FLX/1440209
CN-Du2	GRA	Te	244	9	0.28	0.1	3.0	7.0	0.27	0.4	0.90	(Chen et al., 2019)	https://doi.org/10.18140/FLX/1440140
CZ-BK1	ENF	Te	927	6	0.40	0.2	1.9	10.3	0.35	11.4	1.80	(Acosta et al., 2013)	https://doi.org/10.18140/FLX/1440143
DE-Lnf	DBF	Te	1284	6	0.31	0.32	2.3	4.6	0.37	34.5	1.69	(Anthoni et al., 2004)	https://doi.org/10.18140/FLX/1440150
DE-Tha	ENF	Te	3803	4	0.50	0.1	2.4	7.9	0.31	28.2	2.22	(Prescher et al., 2010)	https://doi.org/10.18140/FLX/1440152
DK-Sor	DBF	Bo	1849	6	0.37	0.1	1.8	5.8	0.38	27.7	1.91	(Pilegaard et al., 2011)	https://doi.org/10.18140/FLX/1440155
ES-LJu	OSH	Me	2838	9	0.55	0.15	2.2	17.6	0.10	0.5	0.38	(Serrano-Ortiz et al., 2007)	https://doi.org/10.18140/FLX/1440157
FI-Hyy	ENF	Bo	3607	3	0.51	0.16	1.8	5.5	0.39	15.0	1.88	(Bäck et al., 2012)	https://doi.org/10.18140/FLX/1440158
FR-LBr	ENF	Te	428	3	0.32	0.3	2.7	10.9	0.20	20.2	2.07	(Berbigier et al., 2001)	https://doi.org/10.18140/FLX/1440163
GF-Guy	EBF	Tr	2903	12	0.33	0.1	4.3	15.7	0.54	35.0	2.83	(Bonal et al., 2008)	https://doi.org/10.18140/FLX/1440165
IT-CA3	DBF	Me	722	6	0.29	0.15	2.1	7.7	0.25	2.2	1.05	(Sabbatini et al., 2016)	https://doi.org/10.18140/FLX/1440232
IT-Col	DBF	Te	1372	6	0.68	0.2	2.7	8.9	0.23	21.5	1.93	(Valentini et al., 1996)	https://doi.org/10.18140/FLX/1440167
IT-Cpz	EBF	Me	719	1	0.19	0.3	2.4	11.7	0.20	14.1	2.59	(Reichstein et al., 2002)	https://doi.org/10.18140/FLX/1440168
IT-Isp	DBF	Te	428	3	0.49	0.5	2.5	17.2	0.31	26.0	1.98	(Ferréa et al., 2012)	https://doi.org/10.18140/FLX/1440234
IT-MBo	GRA	Te	1760	6	0.70	0.2	2.4	9.9	0.33	0.3	1.93	(Marcolla et al., 2011)	https://doi.org/10.18140/FLX/1440170
IT-Noe	CSH	Me	3508	9	0.77	0.3	2.8	7.6	0.21	1.1	1.17	(Marras et al., 2011)	https://doi.org/10.18140/FLX/1440171
IT-PT1	DBF	Te	336	3	0.45	0.15	2.8	7.3	0.19	26.6	1.26	(Migliavacca et al., 2009)	https://doi.org/10.18140/FLX/1440172
IT-Ren	ENF	Te	1927	3	0.60	0.15	3.2	5.1	0.65	29.0	2.18	(Montagnani et al., 2009)	https://doi.org/10.18140/FLX/1440173
IT-Ro2	DBF	Me	1236	10	0.42	0.3	3.0	9.4	0.14	13.5	2.65	(Tedeschi et al., 2006)	https://doi.org/10.18140/FLX/1440175
IT-SRo	ENF	Me	3165	1	0.32	0.4	2.2	11.6	0.22	18.7	1.83	(Chiesi et al., 2005)	https://doi.org/10.18140/FLX/1440176
IT-Tor	GRA	Te	915	3	0.65	0.3	2.8	5.1	0.50	0.2	1.84	(Galvagno et al., 2013)	https://doi.org/10.18140/FLX/1440237
NL-Loo	ENF	Te	4682	2	0.18	0.25	1.6	4.3	0.48	17.0	1.81	(Moors, 2012)	https://doi.org/10.18140/FLX/1440178
RU-Fyo	ENF	Bo	795	3	0.93	0.1	2.4	5.8	0.36	25.4	1.96	(Kurbatova et al., 2008)	https://doi.org/10.18140/FLX/1440183

Table S1 Selected FLUXNET2015 sites characteristics and references (continued)

Site ID	IGBP	Climate	DD	Soil	n	Z_m	E_0	α	λ	h_c	LAI	Reference	Data DOI
US-ARc	GRA	Te	366	4	0.35	0.1	4.2	11.4	0.23	0.4	1.56	(Fischer et al., 2012)	https://doi.org/10.18140/FLX/1440065
US-Blo	ENF	Me	2481	7	0.43	0.1	2.8	16.1	0.23	2.6	2.19	(Goldstein et al., 2000)	https://doi.org/10.18140/FLX/1440068
US-KS2	CSH	Te	1431	1	0.41	0.1	3.2	16.2	0.20	1.5	2.97	(Powell et al., 2006)	https://doi.org/10.18140/FLX/1440075
US-Me2	ENF	Me	3545	3	0.47	0.1	2.9	6.3	0.17	15.7	1.91	(Thomas et al., 2009)	https://doi.org/10.18140/FLX/1440079
US-Me6	ENF	Me	979	3	0.24	0.1	2.8	4.3	0.21	5.9	0.79	(Ruehr et al., 2012)	https://doi.org/10.18140/FLX/1440099
US-MMS	DBF	Te	3421	4	0.62	0.15	2.7	11.6	0.29	26.0	1.82	(Schmid et al., 2000)	https://doi.org/10.18140/FLX/1440083
US-Syv	MF	Te	1063	3	0.32	0.1	2.6	7.5	0.31	27.5	1.95	(Desai et al., 2008)	https://doi.org/10.18140/FLX/1440091
US-Ton	WSA	Me	3146	6	0.55	0.2	3.3	10.0	0.12	9.9	1.21	(Baldocchi et al., 2004)	https://doi.org/10.18140/FLX/1440092
US-Var	GRA	Me	1312	4	0.38	0.2	2.2	7.9	0.27	0.3	2.00	(Ryu et al., 2008)	https://doi.org/10.18140/FLX/1440094

International Geosphere-Biosphere Programme (IGBP) biome classification (evergreen needleleaf forest (ENF), evergreen broadleaf forest (EBF), deciduous broadleaf forest (DBF), mixed forest (MF), closed shrubland (CSH), open shrubland (OSH), woody savanna (WSA), grassland (GRA)); climate (temperate (Tr), temperate (Te), mediterranean (Me), boreal (Bo)); DD, number of daily data used; FAO Soil texture classification ID (sand (1), sandy loam (3), silt loam (4), loam (6), sandy clay loam (7), silty clay loam (8), clay loam (9), sandy clay (10), silty clay (11), clay (12)); soil porosity (n , unitless); depth of soil moisture measurement (Z_m , m); average growing season potential evaporation (Priestley and Taylor, 1972) (E_0 , mm d⁻¹); average growing season rainfall intensity (Rodriguez-Iturbe et al., 1984) (α , mm d⁻¹); average growing season rainfall frequency (Rodriguez-Iturbe et al., 1984) (λ , d⁻¹); average canopy height (h_c , m); average growing season leaf area index (LAI , m² m⁻²). We obtain IGBP, climate, soil ID, Z_m , and h_c from FLUXNET2015 metadata and site references. n is taken as the maximum observed volumetric soil water content in the record. Saturated soil hydraulic conductivity ($k_{s,sat}$, m d⁻¹) and pore size distribution (b , unitless) values are based on soil texture (Rawls et al., 1982). For consistency with soil moisture records, we scale b from the literature values, to constrain the observed daily minimum soil water potential above -10 MPa if that threshold is exceeded in a record. Soil saturation at the hygroscopic point (s_h , unitless) is the soil saturation value at -10 MPa. Soil saturation at field capacity (s_{fc} , unitless) is the maximum between the soil saturation value at -0.03 MPa and the 95th percentile soil saturation value of days following a rain event in each record. Model soil depth parameter (Z , m) is taken as 1.5 times Z_m . We calculate growing season average climate parameters E_0 , α , and λ from FLUXNET2015 daily data. We consider only cloudless LAI data flagged as high-quality between 2003-2013, then interpolate and average values by day of year, using a 30-day convolution to obtain a smooth annual pattern before identifying growing season periods and calculating the average growing season LAI .

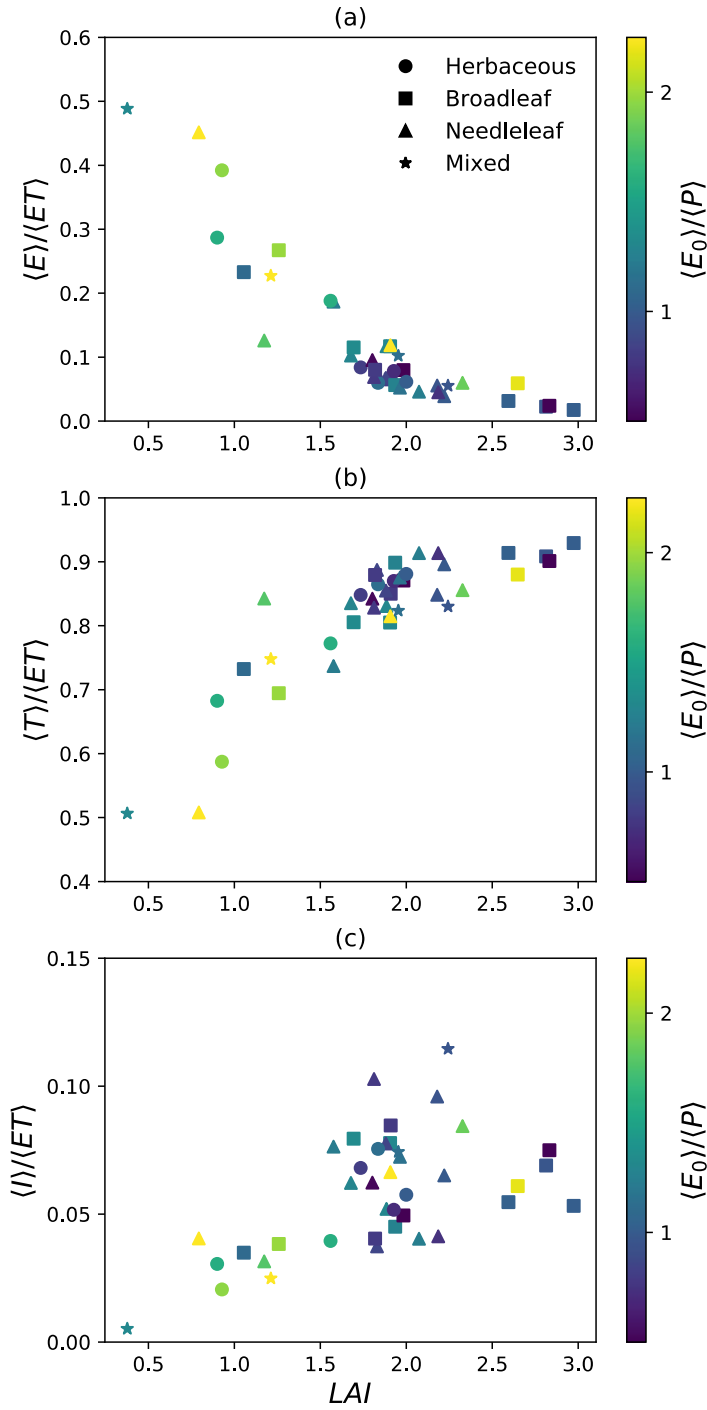


Figure S1 Modeled long-term average growing season total evapotranspiration (ET) partitioning versus leaf area index (LAI). (a) Fraction of soil water evaporation (E). (b) Fraction of transpiration (T). Fraction of canopy interception (I). Marker shapes indicate plant functional type and are colored by the aridity index measured by the ratio between long-term potential evaporation and rainfall ($\langle E_0 \rangle / \langle P \rangle$).

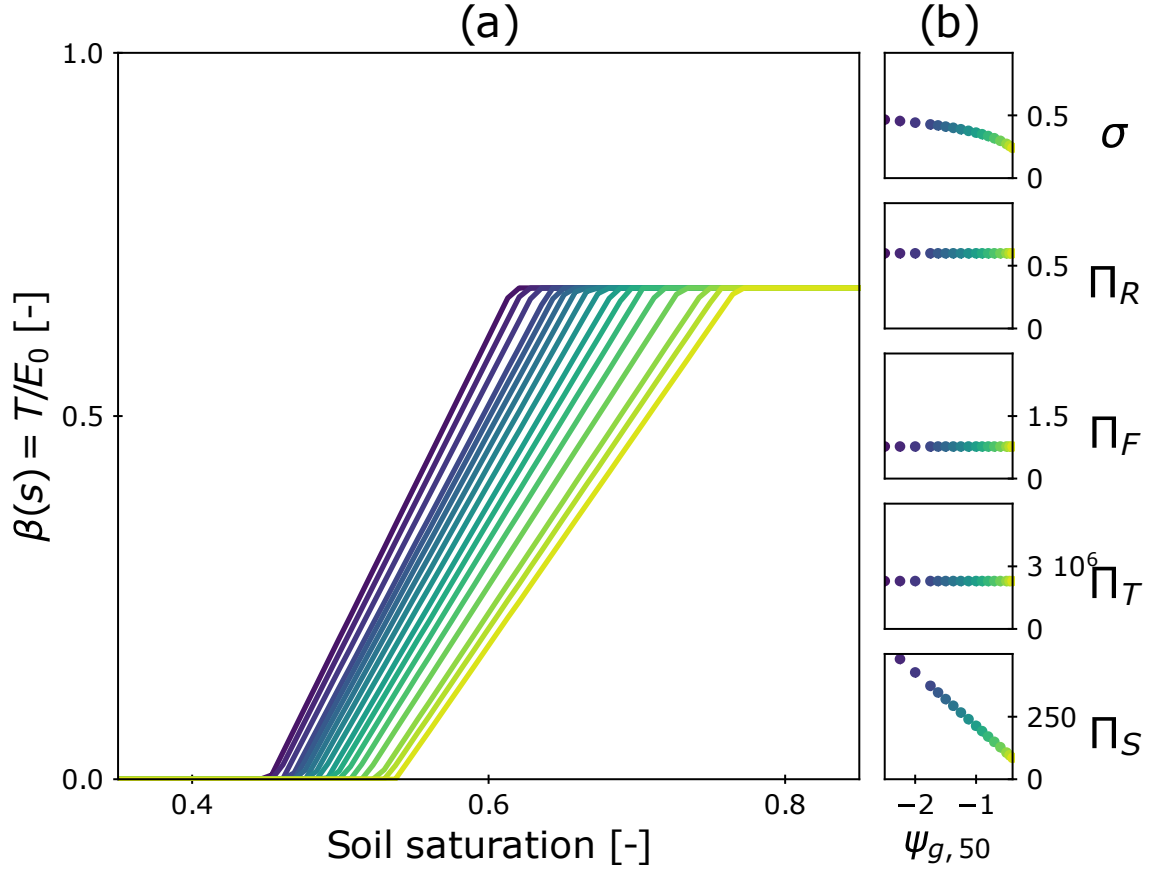


Figure S2 Effect of soil suitability (Π_S) on (a) the fraction of transpiration (T) to potential atmospheric demand (E_0) as a function of soil saturation ($\beta(s)$), (b) plant water uptake capacity (σ) representing the normalized area of the $\beta(s)$ function; and non-dimensional groups for hydraulic risk tolerance (Π_R), plant water flux control (Π_F), soil-to-root water transport capacity (Π_T), and soil suitability (Π_S). We obtain water use strategies for different values of Π_S by varying the water potential at 50% loss of stomatal conductance ($\psi_{g,50}$) from -0.4 to -2.5 MPa (yellow to purple) MPa and maintaining other 3 Π groups constant by adjusting Π_R with the water potential at 50% loss of xylem conductance ($\psi_{x,50}$), Π_F with maximum plant xylem conductance ($K_{P,max}$), and Π_T with maximum soil-root conductance ($K_{SR,max}$) to counterbalance the effect of varying $\psi_{g,50}$. Baseline parameter values are in Table 1.

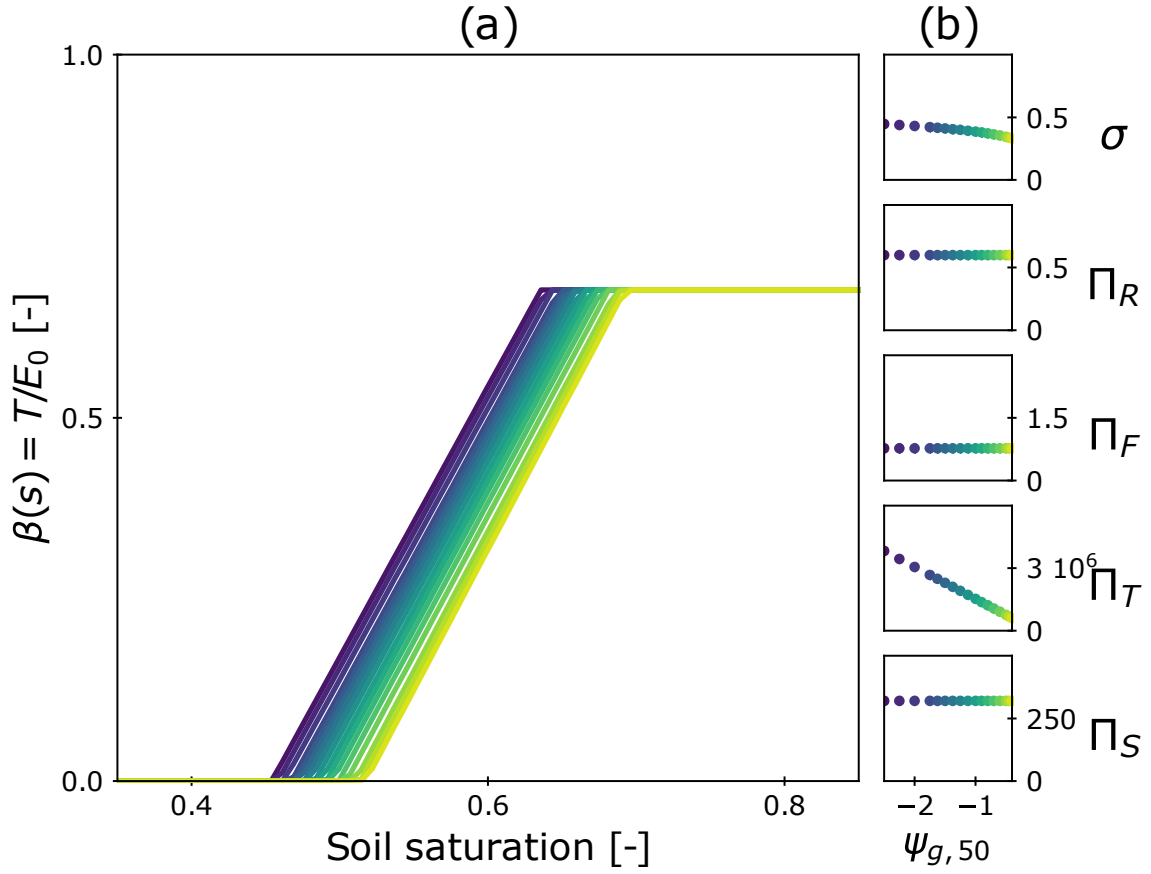


Figure S3 Effect of soil-root transport capacity (Π_T) on (a) the fraction of transpiration (T) to potential atmospheric demand (E_0) as a function of soil saturation ($\beta(s)$), (b) plant water uptake capacity (σ) representing the normalized area of the $\beta(s)$ function; and non-dimensional groups for hydraulic risk tolerance (Π_R), plant water flux control (Π_F), soil-to-root water transport capacity (Π_T), and soil suitability (Π_S). We obtain water use strategies for different values of Π_T by varying the water potential at 50% loss of stomatal conductance ($\psi_{g,50}$) from -0.4 to -2.5 (yellow to purple) MPa and maintaining other 3 Π groups constant by adjusting Π_R with the water potential at 50% loss of xylem conductance ($\psi_{x,50}$), Π_F with maximum plant xylem conductance ($K_{p,max}$), and Π_S with soil water potential near saturation ($\psi_{s,sat}$) to counterbalance the effect of varying $\psi_{g,50}$. Baseline parameter values are in Table 1.

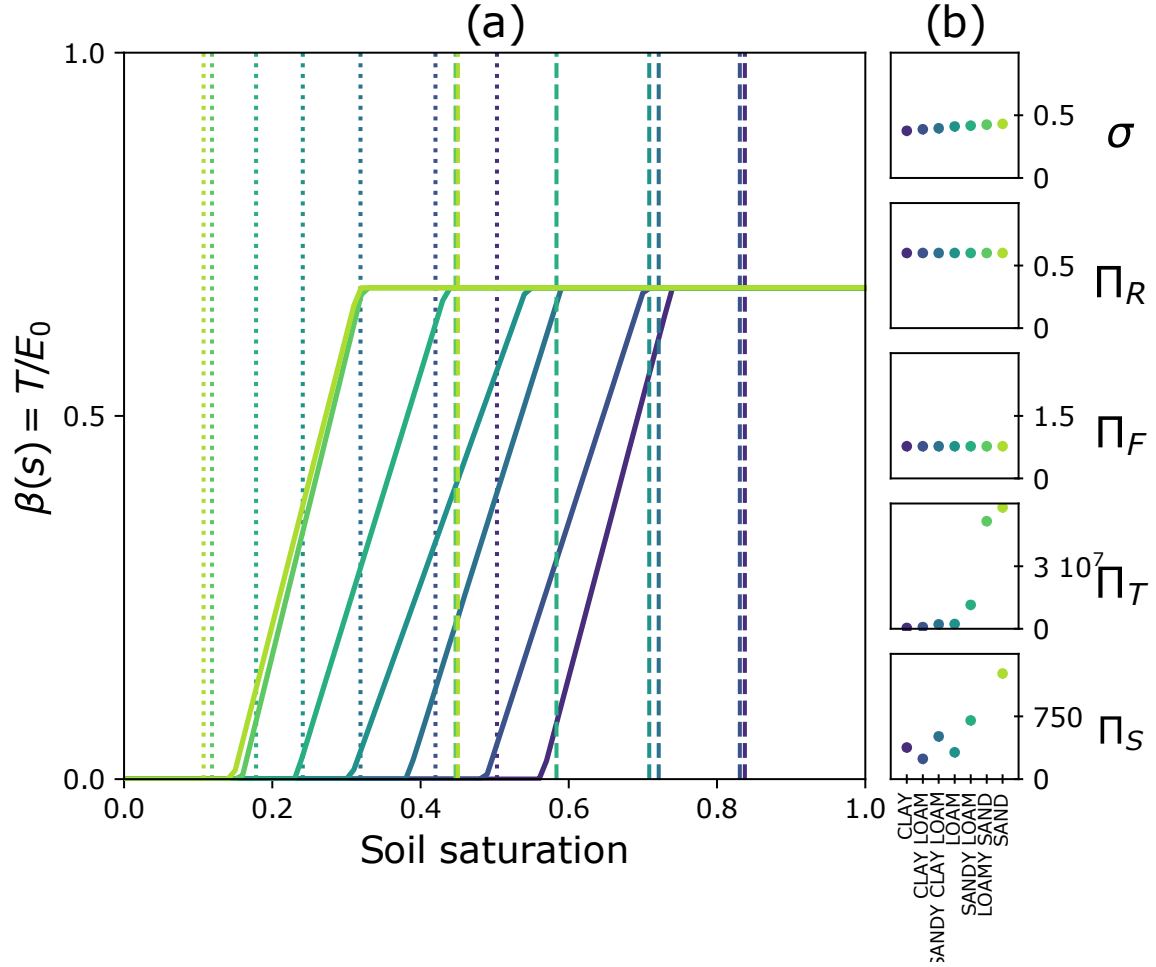


Figure S4 Effect of soil texture on (a) the fraction of transpiration (T) to potential atmospheric demand (E_0) as a function of soil saturation ($\beta(s)$), (b) plant water uptake capacity (σ) representing the normalized area of the $\beta(s)$ function; and non-dimensional groups for hydraulic risk tolerance (Π_R), plant water flux control (Π_F), soil-to-root water transport capacity (Π_T), and soil suitability (Π_S). We vary 4 parameters together, including soil porosity, soil water potential near soil saturation, soil hydraulic conductance, and soil pore distribution for each soil texture (Rawls et al., 1982) from most to least draining soils (yellow to purple). We calculate soil saturation at the hygroscopic point (s_h , dotted line) at -10 MPa and at field capacity (s_{fc} , dashed line) at -0.03 MPa based on soil texture and use other climate and vegetation parameter values in Table 1.

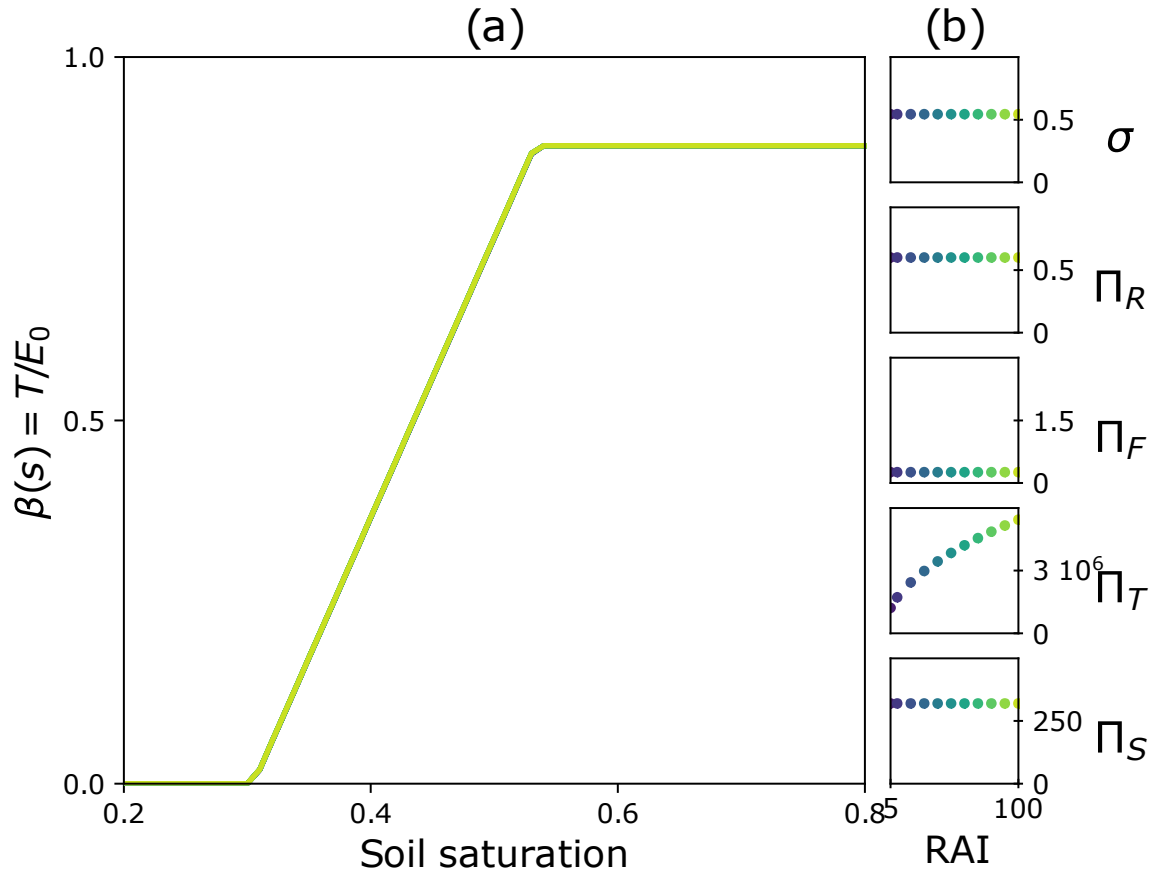


Figure S5 Effect of root area index (RAI) on (a) the fraction of transpiration (T) to potential atmospheric demand (E_0) as a function of soil saturation ($\beta(s)$), (b) plant water uptake capacity (σ) representing the normalized area of the $\beta(s)$ function; and non-dimensional groups for hydraulic risk tolerance (Π_R), plant water flux control (Π_F), soil-to-root water transport capacity (Π_T), and soil suitability (Π_S). We obtain water use strategies with root area index (RAI) ranging from 5 to 100 $\text{m}^2 \text{ m}^{-2}$ (yellow to purple) and baseline parameter values in Table 1, except we multiply xylem conductance by 3 to test limitations of soil-root conductance in a more extreme case.

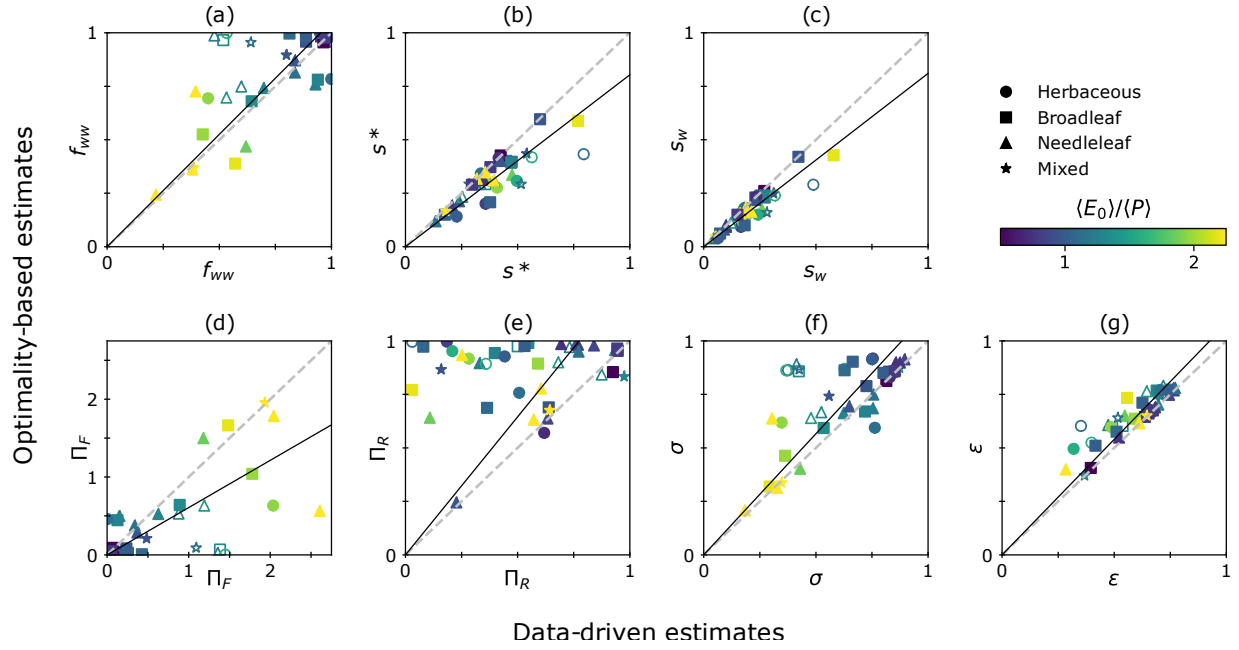


Figure S6 Same as Figure 9 in main text but considering static versus dynamic plant water stress. Relation between data-driven water use strategies that maximizes the goodness-of-fit between empirical and theoretical $p(s)$ and optimality-based estimates that maximize water use performance using data from study sites. (a) fractional loss of conductance in well-watered conditions (f_{ww} , $r = 0.73$, RMSD = 0.18, bias = 7.8 %); (b) soil saturation at the point of incipient stomatal closure (s^* , $r = 0.84$, RMSD = 0.10, bias = -20.7 %); (c) soil saturation at the wilting point (s_w , $r = 0.92$, RMSD = 0.06, bias = 19.9 %); (d) plant water transport control (Π_F , $r = 0.79$, RMSD = 0.62, bias = -64.6 %); (e) plant hydraulic risk tolerance (Π_R , $r = \text{non-significant}$, RMSD = 0.46, bias = 39.6 %); (f) plant water uptake capacity (σ , $r = 0.66$, RMSD = 0.20, bias = 16.1 %); (g) plant water use performance (ε , $r = 0.90$, RMSD = 0.08, bias = 8.8 %). Markers indicate ecosystem dominant plant functional type. Mixed ecosystems include temperate and boreal forests with mixture of broadleaf and needleleaf trees as well as tropical and Mediterranean savannas and shrublands with a mixture of herbaceous and woody vegetation. Color indicates aridity, measured by the ratio between long-term potential evaporation (E_0) and rainfall (P). Markers with empty face colors distinguish sites with poor goodness of fit (NSE<0) between the theoretical and empirical using the optimality-based parameter estimates (5 out of 40) and 2 sites are not represented because they did not lead to a converging result. Grey dashed lines indicate the 1:1 lines.

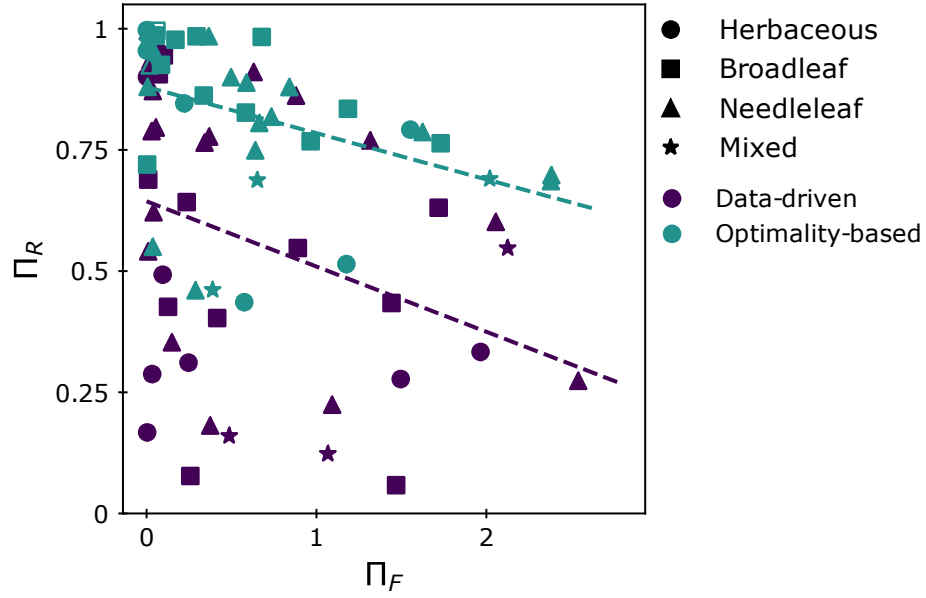


Figure S7 Relation between inferred plant water flux control (Π_F) and hydraulic risk tolerance (Π_R) from soil moisture data (dark purple) and an optimality criteria (light blue) at study sites. Marker shapes indicate plant functional type. Dashed lines represent the fitted trend for both data-driven ($slope = -0.13$; $p\text{-value} = 0.03$) and optimality-based ($slope = -0.10$, $p\text{-value} = 0.01$) estimates.

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